

Equation of Straight Line

$$ax + by + c = 0$$

$$by + c = -ax$$

$$by = -ax + (-c)$$

$$y = \frac{-a}{b}x + \frac{-c}{b}$$

Equation of Straight Line

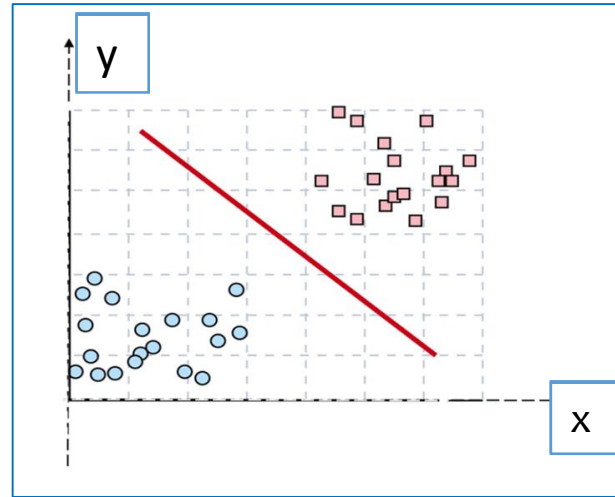
$$y = m x + c$$

m OR $\frac{-a}{b}$ = (Slope or Coefficient) slope of the line

c OR $\frac{-c}{b}$ = Intercept (where the best-fit line meets the Y axis).

Equation of Straight Line

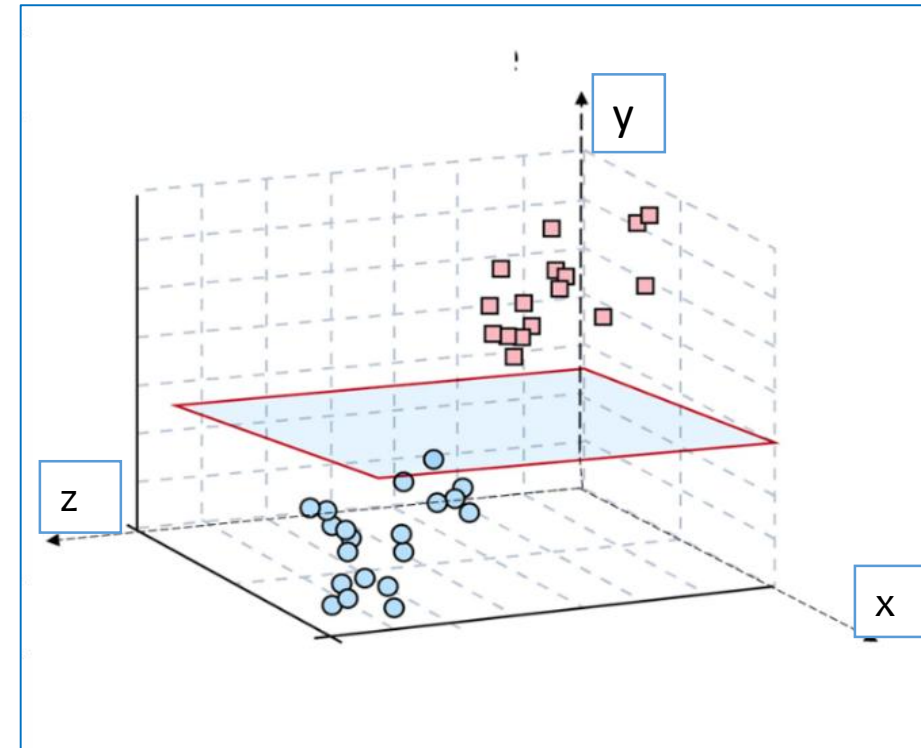
$$ax + by + c = 0$$



Equation of Straight Line 3 Dimension

$$ax + by + cz + c = 0$$

$$ax_1 + bx_2 + cx_3 + c = 0$$



Equation of Straight Line

$$ax + by + cz + c = 0$$

$$ax_1 + bx_2 + cx_3 + c = 0$$

If $w_1 = a$

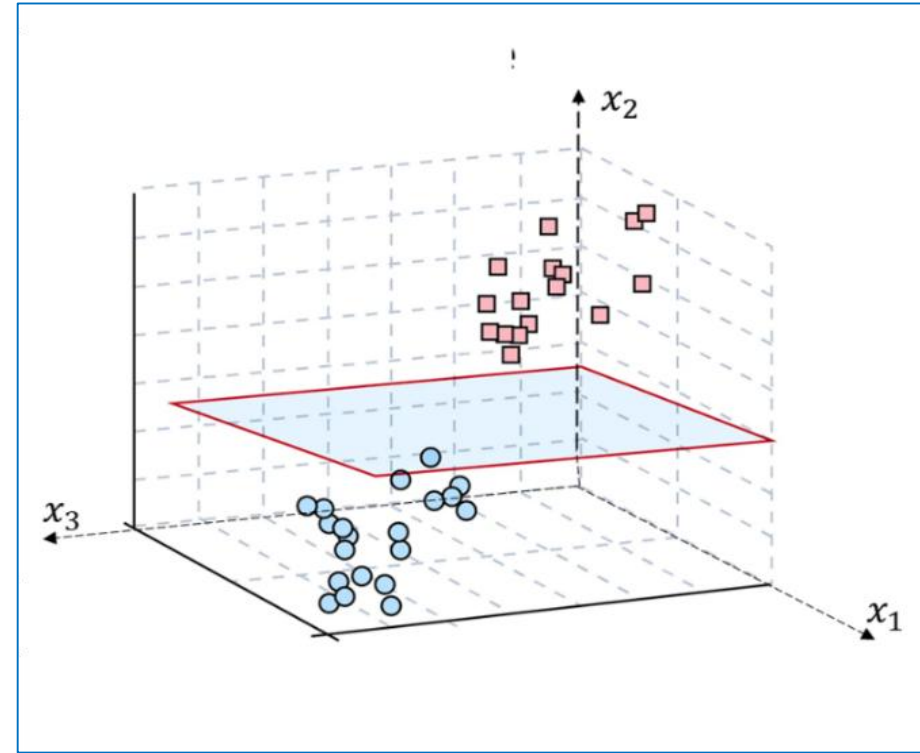
$$w_2 = b$$

$$w_3 = c$$

$$w_1x_1 + w_2x_2 + w_3x_3 + c = 0$$

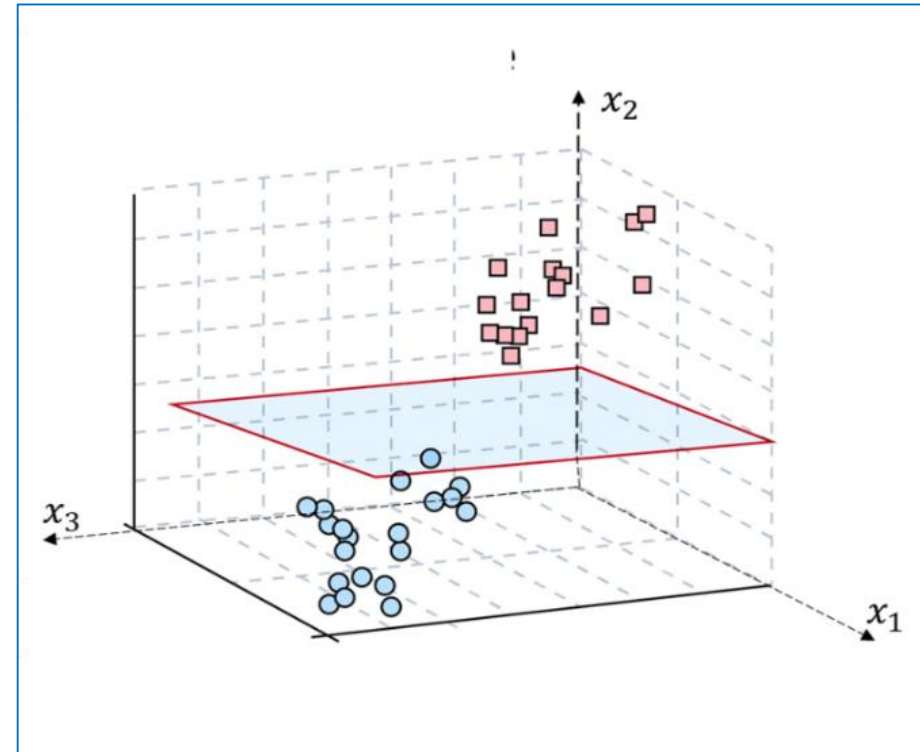
$$w_1x_1 + w_2x_2 + w_3x_3 + c = 0$$

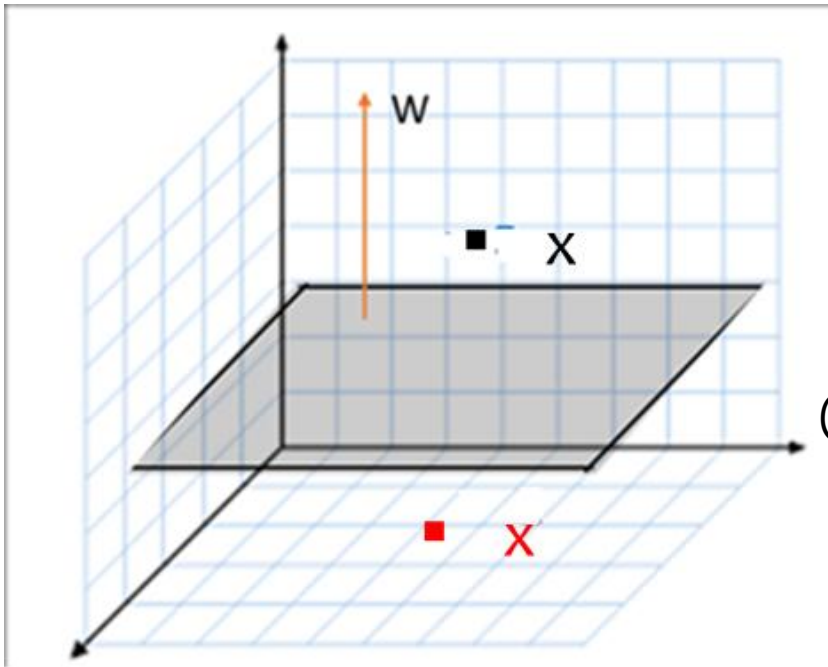
$$w^T x + c = 0$$



$$w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 \dots + w_nx_n + c = 0$$

$$w^Tx + c = 0$$



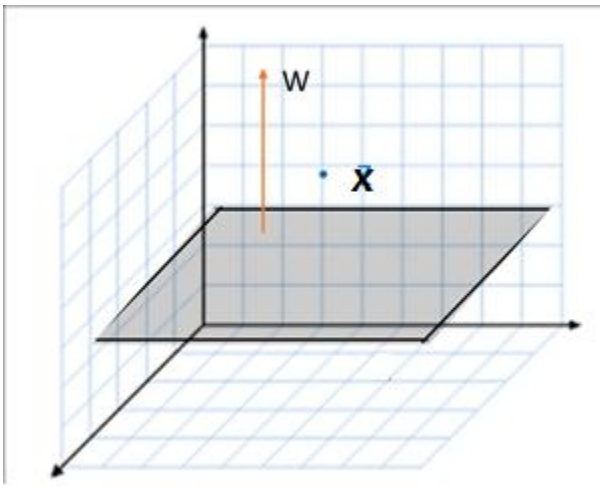


Consider w is a vector which is perpendicular to the plane and x situated above and below the plane respectively. If point lies on the side of the vector it means it is +ve points and vice versa.

Using the below equation, we can determine the position of the points (values) relative to the plane, whether it lies above or below the hyperplane.

$$||w|| \ ||x|| \ \cos\theta = +d$$

$$||w|| \ ||x|| \ \cos\theta = -d$$

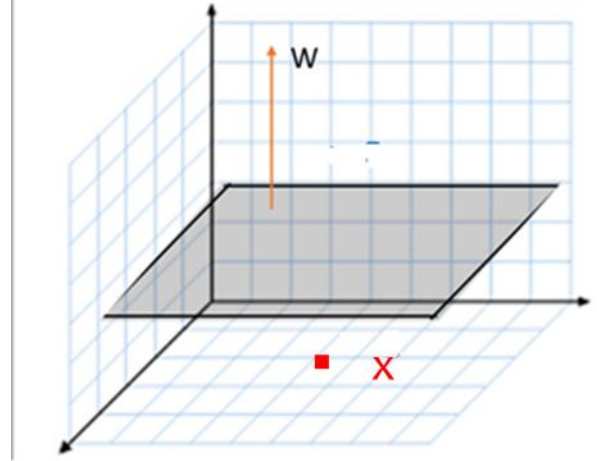


$$d = \frac{w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0}{||w||}$$

$$d = \frac{w^T x + w_0}{||w||}$$

$$\frac{w^T x + w_0}{||w||} = +d$$

$$||w|| \quad ||x|| \cos \theta = +d$$

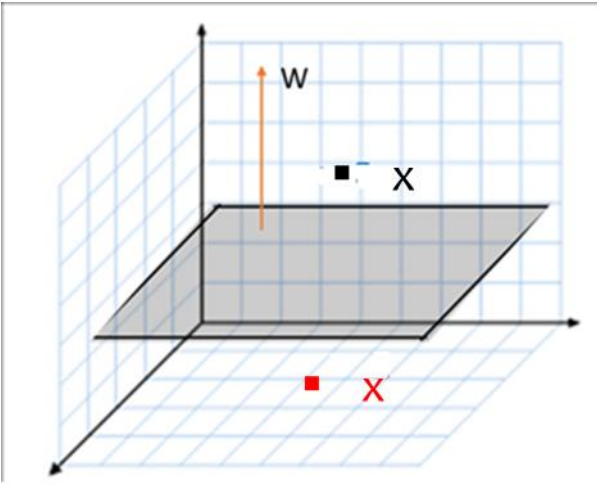


$$d = \frac{w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0}{||w||}$$

$$d = \frac{w^T x + w_0}{||w||}$$

$$\frac{w^T x + w_0}{||w||} = -d$$

$$||w|| \quad ||x|| \cos \theta = -d$$



The crucial factor in this equation is the role played by cosine theta ($\cos\theta$), as it helps ascertain the point's position based on the angle it forms with the vector.

- Therefore, when applying the $\cos\theta$ value in the equation,
- points with an angle **less** than 90 degrees yield a **positive value**,

$$||w|| \ ||x|| \ \cos\theta = +d$$

$$||w|| \ ||x|| \ \cos\theta = +d$$

- while those with an angle **greater** than 90 degrees result in a **negative value**.

$$||w|| \ ||x|| \ \cos\theta = -d$$